SK Khalic Meera

28 Jul 2002 | Narsipatnam, Andhra Pradesh, India

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Education

National Institute of Technology, Rourkela — **2025 – Present**
M.Tech in Communication and Networks ***(CGPA: 8.33)*** — *Rourkela, Odisha*

Koneru Lakshmaiah University — **2019 – 2023**
B.Tech in Computer Science and Engineering ***(CGPA: 8.40)*** — *Vijayawada, Andhra Pradesh*

Bhashyam Junior College — **2017 – 2019**
Intermediate ***(CGPA: 9.20)*** — *Kakinada, Andhra Pradesh*

Bhashyam Public School — **2016 – 2017**
SSC ***(CGPA: 10.00)*** — *Tuni, Andhra Pradesh*

Relevant Coursework

- Embedded Systems
- Computer Networks
- Digital Electronics
- Operating Systems
- Computer Architecture
- Data Structures & Algorithms

Experience

Technical Intern – Mobile Application Development — **Jul 2022 – Dec 2022**
FUTZEN E-Technologies Pvt. Ltd.

- Developed Android applications using **Java** and **XML** in Android Studio.
- Designed and maintained **MySQL** databases for backend data management.

Projects

HTTP Web Server | C++, Linux, TCP/IP, HTTP — **Jul 2026**

- Developed a lightweight HTTP web server in **C++** using **Linux socket programming** and **TCP/IP**.
- Implemented an event-driven architecture using **epoll** to handle multiple concurrent client connections efficiently.
- Designed a **thread pool** for concurrent request processing and efficient static file serving.

Smart Traffic Light Controller | ESP32, C++, FreeRTOS — **May 2026**

- Developed a real-time traffic light controller on **ESP32** using **C++** and **FreeRTOS**.
- Implemented concurrent tasks for traffic signal control, vehicle detection, and pedestrian crossing management.
- Used **FreeRTOS queues**, **semaphores**, and **GPIO** for task synchronization and hardware interfacing.

AI-Powered Document Intelligence Assistant | Python, RAG, FAISS, FastAPI — **Jul 2026**

- Developed a **RAG**-based document question-answering application using **Python** and LLM APIs.
- Implemented document ingestion, text chunking, embedding generation, and semantic retrieval using **FAISS**.
- Built modular **FastAPI** endpoints for document processing and question answering.

Technical Skills

- **Programming:** C, C++, Python
- **Systems & Concurrency:** Linux, Multithreading, Thread Pools, Synchronization, FreeRTOS
- **Networking:** TCP/IP, HTTP, Socket Programming, Computer Networks
- **Embedded Systems:** ESP32, GPIO, SPI, I2C, PWM
- **AI/GenAI:** RAG, LLM APIs, Embeddings, FAISS, Semantic Retrieval

Certifications & Achievements — View Folder

- **Paper Presenter** – 2026 IEEE 1st International Conference on Instrumentation (INSTCon 2026), presented the paper *"Machine Learning-Based Real-time Intrusion Detection System for IoMT Networks"*.
- **1st Prize** – IEEE WIE Day 2026 Idea Pitch & Poster Presentation for *"ML-Based Intrusion Detection System for Internet of Medical Things"*.